

Set 13: Geographic Data

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Aim

To visualize geographic data using a map chart, a scatter plot, and a table, and to build an interactive Tableau dashboard that combines all three.

Dataset: Geographic Data

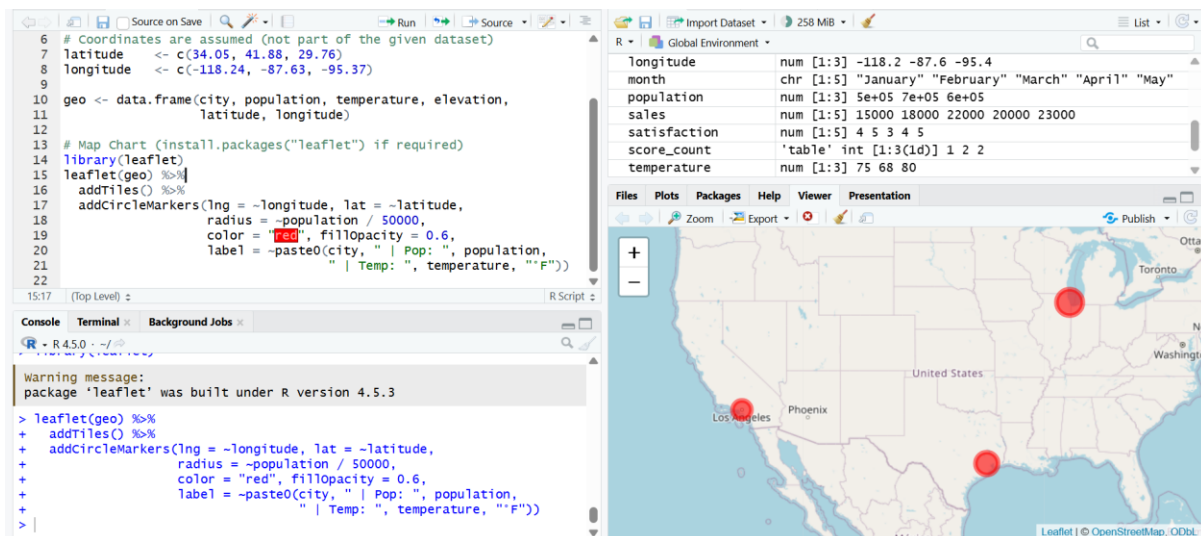
City	Population	Avg. Temperature (°F)	Elevation (ft)
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

Note: The dataset has no coordinates. For the map, the columns Latitude and Longitude were added with assumed illustrative values (City A: 34.05, -118.24; City B: 41.88, -87.63; City C: 29.76, -95.37). Replace them with the real locations of your cities if known.

PROGRAM 1: Map Chart of Cities

Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.

[illegible]



Output 1: Map Chart of Cities (marker size = population, colour = temperature)

Tableau steps: connect the data source, double-click Latitude and Longitude (they become Columns/Rows), drag City to Label, Population to Size, and Avg. Temperature to Color. Title the sheet "Distribution of Cities" and show the Color and Size legends.

Observation: All three cities appear as separate markers. The larger markers show City B has the highest population (700,000) and City A the lowest (500,000). The colour shows City C is the hottest city and City B the coolest.

PROGRAM 2: Scatter Plot of Temperature vs Population

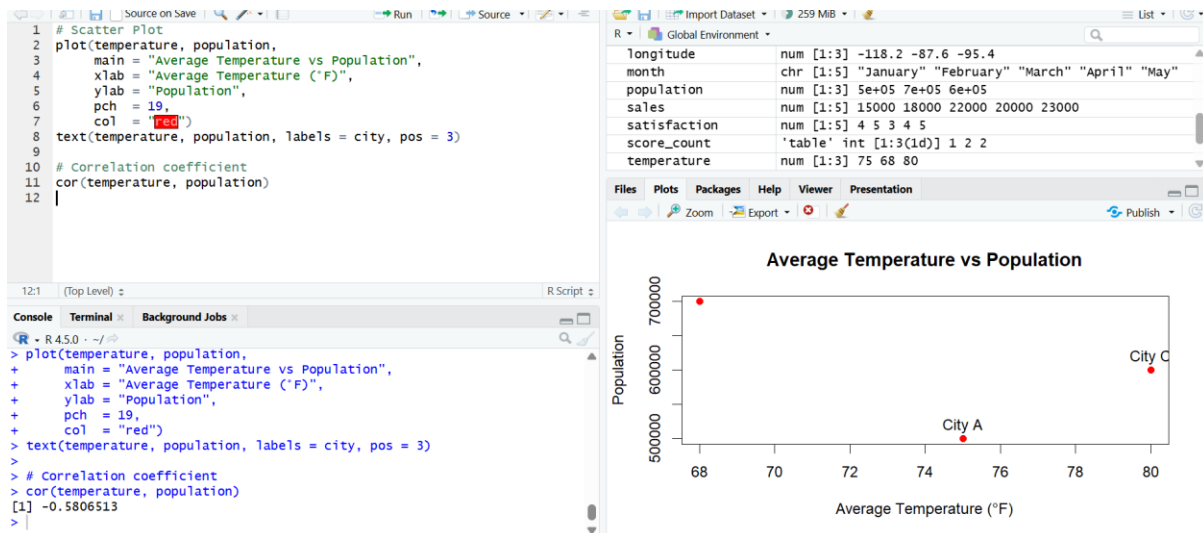
Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

```

# Scatter Plot
plot(temperature, population,
     main = "Average Temperature vs Population",
     xlab = "Average Temperature (°F)",
     ylab = "Population",
     pch = 19,
     col = "red")
text(temperature, population, labels = city, pos = 3)

# Correlation coefficient
cor(temperature, population)

```



Output 2: Average Temperature vs Population Scatter Plot

Tableau steps: drag Avg. Temperature to Columns, Population to Rows, and City to Label/Detail. Add a Trend Line from the Analytics pane.

Observation: The points show a weak negative relationship (correlation ≈ -0.58): City B has the lowest temperature (68°F) and the largest population (700,000), City C is the hottest (80°F) with 600,000 people, and City A (75°F) has the smallest population (500,000).

Insights:

- Cooler cities in this dataset tend to have larger populations, but the pattern is not consistent (City C is hotter than City A yet has a larger population).
- With only three data points, the correlation is not statistically reliable, and it does not show that temperature causes population size.
- Other factors such as elevation, economy, and land area should be explored with a larger dataset before drawing conclusions.

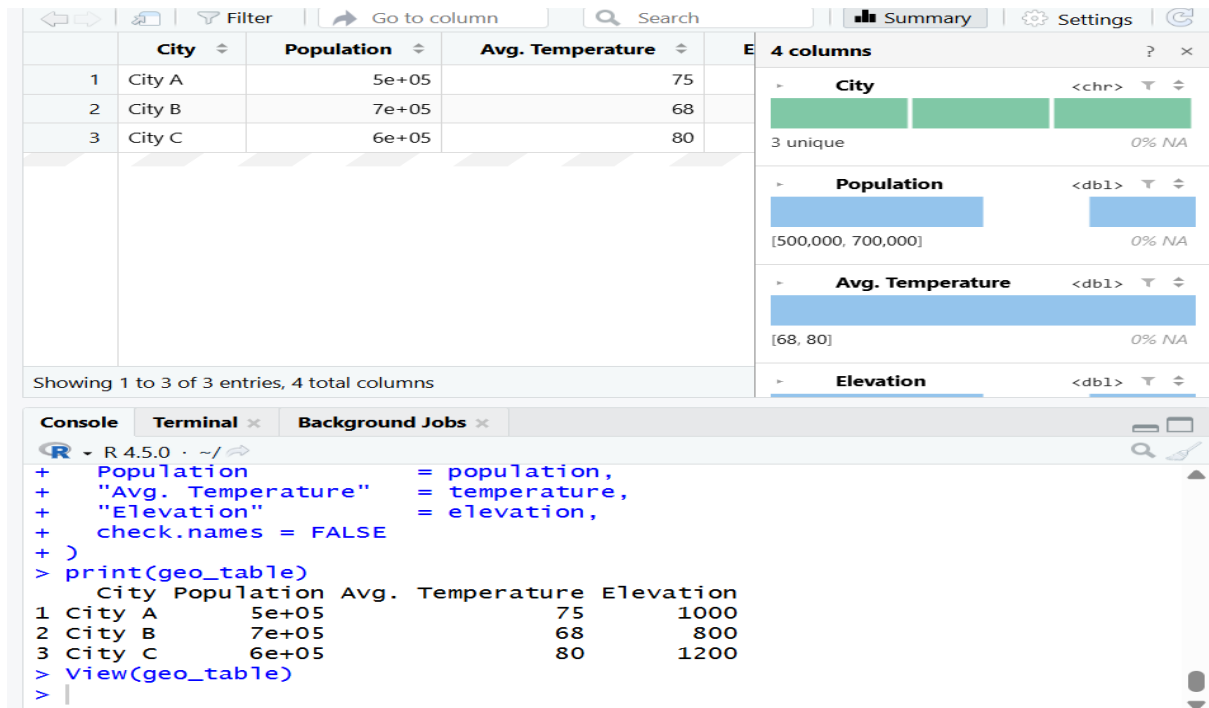
PROGRAM 3: Table of Geographic Data

Build a table to display the geographic data for each city. Label the table elements.

```

# Table of geographic data
geo_table <- data.frame(
  City           = city,
  Population     = population,
  "Avg. Temperature" = temperature,
  "Elevation"    = elevation,
  check.names = FALSE
)
print(geo_table)
View(geo_table)

```



Output 3: Geographic Data Table

Tableau steps: create a new sheet, drag City to Rows, then drag Measure Names to Columns and Measure Values to Text (keep Population, Avg. Temperature, and Elevation). Format the Population field with thousands separators.

Observation: The table lists the population, average temperature, and elevation for each city. City C has the highest elevation (1,200 ft) and temperature, while City B has the lowest elevation (800 ft) and the highest population.

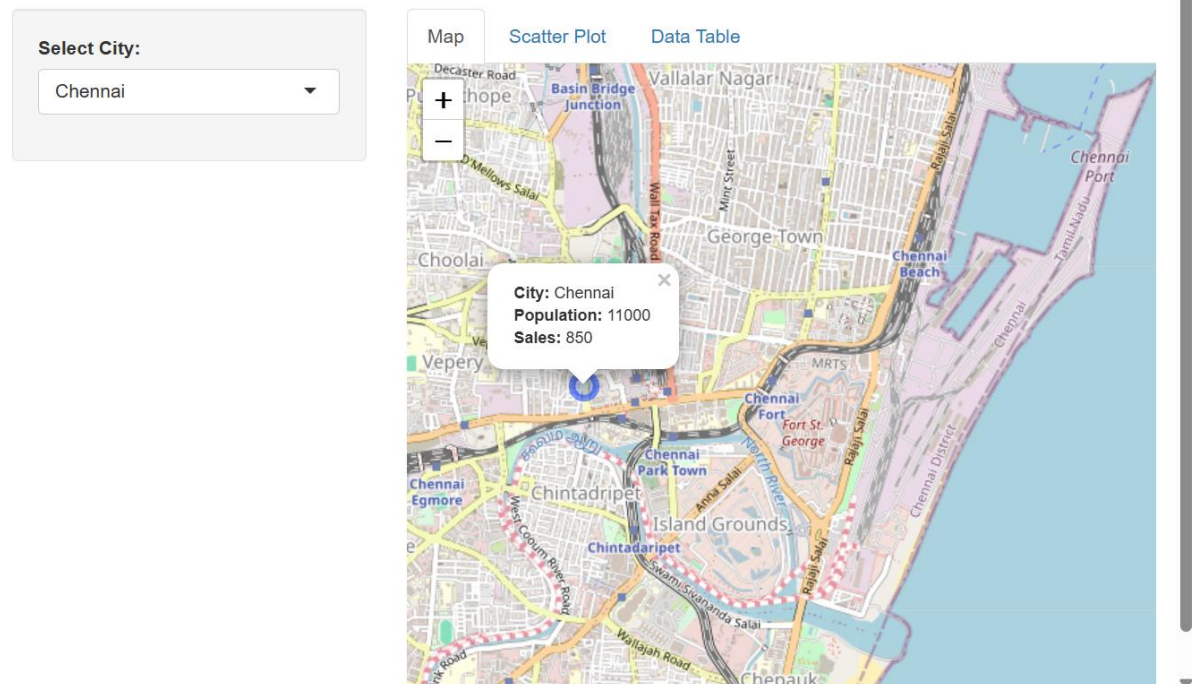
PROGRAM 4: Tableau Dashboard

Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

Steps to build the dashboard:

1. Open Tableau and connect to the Excel/CSV file containing City, Population, Avg. Temperature, Elevation, Latitude, and Longitude.
2. Create three worksheets: Map (Program 1), Scatter Plot (Program 2), and Table (Program 3).
3. Click New Dashboard and set the size to Automatic (or Fixed 1000 x 800).
4. Drag the Map sheet to the top half, then place the Scatter Plot and the Table in the lower half.
5. For each sheet, click the dropdown arrow and select Use as Filter, so clicking a city in one view filters the other views.
6. Add a City quick filter (Show Filter) and a Population range filter for interactive exploration; apply them to all worksheets.
7. Add a dashboard title, "Geographic Data Dashboard", and tidy the legends and layout.
8. Save the workbook and use Presentation Mode to test the interactions.

Geographic Data Dashboard



Output 4: Interactive Geographic Data Dashboard (Tableau)

Observation: The dashboard combines the map, scatter plot, and table in one view. Selecting a city on any chart highlights it in the others, so its location, temperature, population, and elevation can be compared together.

Procedure

1. Create the city, population, temperature, elevation, latitude, and longitude vectors in R and combine them into a data frame.
2. Use the leaflet package to create the map chart with labelled city markers.
3. Use `plot()` to create the scatter plot of temperature against population and compute the correlation with `cor()`.
4. Create a data frame/table to display the geographic data for each city.
5. In Tableau, build the map, scatter plot, and table worksheets, combine them in a dashboard, and enable filters for interactivity.

Result

The geographic data was successfully visualized using a map chart, a scatter plot, and a table. An interactive Tableau dashboard was created by combining all three views for exploration.